

EDA.ipynb – Technical Report

Exploratory Data Analysis Notebook Review — Loan Default Prediction (XAI) Project

1. Purpose of This File in the Project

EDA.ipynb represents the **first stage of the machine learning pipeline** for the Loan Default Prediction / XAI project. Its role is to build an evidence-based understanding of the raw dataset before any model (e.g. the LightGBM / XGBoost models used in `backend_api`) is trained on it. Without this step, feature selection, data cleaning decisions, and model evaluation strategy would be made blindly. Specifically, this notebook is meant to answer:

- Is the dataset clean (missing values, duplicates)?
- Is the target variable (Default) imbalanced?
- Which features correlate most strongly with loan default?
- Are there relationships or outliers that need to be handled before training?

2. Dataset Overview

Source: **Kaggle – Loan Default Dataset** (kaggle.com/datasets/nikhil1e9/loan-default)

Metric	Value
Rows	255,347
Columns	18
Memory usage	≈ 35.1 MB
Missing values	0 (0.00%) — fully clean
Duplicate rows	0 (0.000%)
Target variable	Default (0 = repaid, 1 = defaulted)
Default rate	≈ 11.6% (imbalanced)

Column Reference

Column	Type	Description / Notes
LoanID	object	Unique identifier (255,347 unique values)
Age	int64	Applicant age — 52 unique values
Income	int64	Annual income
LoanAmount	int64	Requested loan amount

CreditScore	int64	550 unique values
MonthsEmployed	int64	Employment tenure in months
NumCreditLines	int64	Only 4 unique values
InterestRate	float64	Loan interest rate
LoanTerm	int64	Only 5 unique values (months)
DTIRatio	float64	Debt-to-income ratio
Education	object	4 categories
EmploymentType	object	4 categories
MaritalStatus	object	3 categories
HasMortgage	object	Yes / No
HasDependents	object	Yes / No
LoanPurpose	object	5 categories
HasCoSigner	object	Yes / No
Default	int64	Target variable

3. Step-by-Step Walkthrough of the Notebook

Step 1 – Environment Setup

Imports pandas, numpy, matplotlib, seaborn and plotly; configures visual style (seaborn whitegrid theme, plot size, Plotly renderer for Colab).

Step 2 – Load Data & Initial Inspection

Loads `Loan_default.csv` and displays `head()`, `tail()` and `dtypes`. **Issue:** the file path is hard-coded to a local Windows path (`C:\Users\User\Desktop\XAI_System\data\raw\Loan_default.csv`), which will fail on any other machine unless a relative path or environment variable is used.

Step 3 – Data Quality Assessment

Runs `df.info()`, computes missing-value counts/percentages, checks for duplicate rows, and lists the number of unique values per column. Result: the dataset is completely clean — zero missing values and zero duplicates.

Step 4 – Statistical Summary

Uses `describe()` for numerical columns and `value_counts()` for categorical columns. Automatically detects the target column (Default) and visualizes its class distribution as a pie chart.

Steps 5–8 – Univariate / Bivariate / Multivariate / Advanced Visualizations

The notebook's own outline (Pipeline Overview, first cell) promises dedicated sections for univariate distributions, bivariate analysis against the target, a multivariate **correlation heatmap**, and advanced Plotly visuals. In the delivered notebook these section headers and their core cells are missing — the content jumps directly from "Step 4" to ad-hoc 3D scatter plots. See Section 5 (Issues Found) for details.

Partially present – 3D Scatter Plots (Plotly)

Three interactive 3D scatter plots are present, colored by Default status: Age × LoanAmount × CreditScore, Income × MonthsEmployed × InterestRate, and NumCreditLines × LoanTerm × DTIRatio.

Partially present – Categorical Default-Rate Bar Charts

For each of Education, EmploymentType, MaritalStatus, LoanPurpose, HasMortgage, HasDependents and HasCoSigner, the notebook groups by category and plots the mean default rate (%) per category as an interactive bar chart.

Step 9 – Professional Insights & Recommendations

Prints a data-quality summary table, target variable analysis (default rate, class balance assessment, business impact), the top positively/negatively correlated features, and a set of actionable recommendations for risk management, feature engineering, and business strategy (e.g. using SMOTE/class weights, feature interactions, dynamic pricing).

Final Cleanup and Export

Creates a `df_clean` copy intended to hold the final cleaning steps before modeling. **Note:** no actual cleaning/encoding code is implemented here — only a placeholder comment ("Add your cleaning steps here").

4. Key Findings (From Actual Executed Output)

The notebook contains saved outputs from a prior execution. These were inspected directly to confirm real results rather than relying on the code alone:

Finding	Result
Total records	255,347
Total features	18
Missing values	0 (0.00%) — no imputation required
Duplicate rows	0 — no de-duplication required
Default rate	≈ 11.6% → moderately imbalanced target
Strongest reported default indicators	Interest Rate and Age
Portfolio risk classification	Moderate risk (10–20% default range)

Because the default rate sits below 20%, the dataset is imbalanced enough that model training should account for it (e.g. SMOTE, class weighting, or threshold tuning) to avoid a model that simply predicts "no default" for everyone and still scores a high accuracy.

5. Issues Found in the Notebook

The following problems were identified by inspecting both the code and the saved cell outputs. These matter if the notebook is to be re-run, graded, or used as-is:

5.1 Undefined variable breaks the Insights section

The Step 9 cell references a variable named `corr_matrix` (the correlation matrix) to compute the top positively/negatively correlated features with Default. This variable is never created anywhere in the notebook's visible cells. The saved output for that exact cell confirms the notebook actually failed when run:

```
NameError: name 'corr_matrix' is not defined
```

This strongly suggests that a cell producing the correlation heatmap (typically `df.corr() + sns.heatmap()`) was deleted or lost from this copy of the notebook.

5.2 Undefined variable used in 3D plots

The three 3D scatter plot cells and the categorical bar-chart cell all rely on a variable named `numerical_cols`, which is likewise never defined in any visible cell. Their outputs are present (cached from an earlier run in a different notebook state), but running the notebook fresh, top to bottom, would raise the same kind of error here too.

5.3 Missing pipeline sections

The notebook's own introduction promises Steps 5–8 (Univariate, Bivariate, Multivariate, Advanced Visualizations) as dedicated, labeled sections. Those headers and their core cells (individual

feature-distribution histograms, boxplots against the target, and the correlation heatmap itself) do not exist in the delivered file — only some downstream multivariate/bivariate outputs (3D scatter plots, bar charts) survive, without the analysis that was supposed to produce their supporting variables.

5.4 Hard-coded local file path

Data loading uses an absolute Windows path specific to one contributor's machine. This should be replaced with a relative path or a configurable environment variable so the notebook is portable.

5.5 Empty cleaning step

The "Final Cleanup and Export" section copies the dataframe (`df_clean = df.copy()`) but implements no actual cleaning, encoding, or scaling logic, despite the notebook declaring "EDA Complete! Ready for modeling."

5.6 Duplicated/misplaced loading cell

The very last cell of the notebook re-implements data loading using a relative filename with an existence check, apparently as a fallback for the hard-coded path in Step 2. It is placed after the "End of Report" marker, which suggests it was appended later and not properly integrated into the notebook's flow.

6. Recommendations

Fix the correlation step. Add a cell that computes `corr_matrix = df.select_dtypes(include=[np.number]).corr()` and renders it as a heatmap before Step 9 runs.

Define numerical_cols explicitly. e.g. `numerical_cols = df.select_dtypes(include=[np.number]).columns`, placed early in Step 4 so downstream cells (3D plots, insights) can rely on it.

Restore the missing analysis sections. Add univariate histograms/boxplots for each numerical feature and bivariate plots against Default, matching the outline promised at the top of the notebook.

Use a portable data path. Replace the hard-coded Windows path with a relative path or a config/environment variable.

Implement real cleaning logic. Populate the Final Cleanup section with the actual encoding (categorical → numeric), scaling, and any imputation logic actually used before model training.

Address class imbalance explicitly. Given an 11.6% default rate, document and apply a chosen strategy (SMOTE, class weights, or threshold tuning) before/while training the model.

Note: This EDA is exploratory in nature. All insights should be validated with statistical testing and business context before being used to make lending decisions, and correlation should not be interpreted as causation.